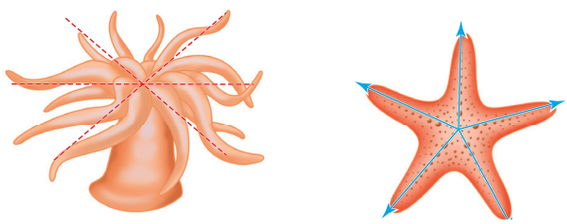
biradially symmetrical (Figure 2.4). Biradial symmetry is a combination of radial and bilateral symmetry as seen in ctenophores. There are only two planes of symmetry, one through the longitudinal and sagittal axis and the other through the longitudinal and transverse axis. (e.g., Comb jellyfish – *Pleurobrachia*)

Animals which have two similar halves on either side of the central plane show **bilateral symmetry** (Figure 2.5). It is an advantageous type of symmetry in triploblastic animals, which helps in seeking food, locating mates and escaping from predators more efficiently. Animals that have dorsal and ventral sides, anterior and posterior ends, right and left sides are bilaterally symmetrical and exhibit cephalisation, in which the sensory and brain structures are concentrated at the anterior end of the animal.



Radial symmetry in sea anemone *Pentamerous radial symmetry in starfish*

Figure. 2.3 Radial and Pentamerous radial symmetry

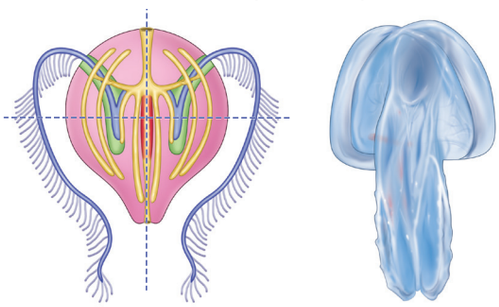


Figure 2.4 Biradial symmetry in comb jelly

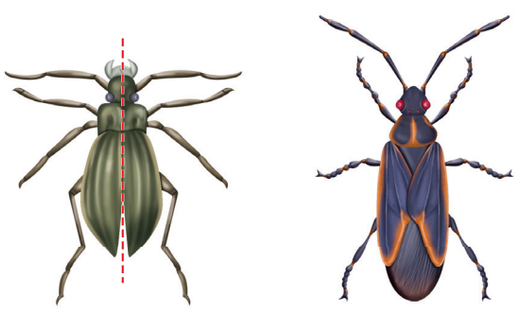


Figure 2.5 Bilateral symmetry in Insects

2.1.4 Coelom

The presence of body cavity or coelom is important in classifying animals. Most animals possess a body cavity between the body wall and the alimentary canal, and is lined with mesoderm (Figure 2.6)

Animals which do not possess a body cavity are called **acoelomates**. Since there is no body cavity in these animals their body is solid without a perivisceral cavity, this restricts the free movement of internal organs. (e.g., Flatworms)

In some animals, the body cavity is not fully lined by the mesodermal epithelium, but the mesoderm is formed as scattered pouches between the ectoderm and endoderm. Such a body cavity is called a **pseudocoel** and is filled with pseudocoelomic fluid. Animals that possess a pseudocoel are called **pseudocoelomates** e.g., Round worms. The pseudocoelomic fluid in the pseudocoelom acts as a hydrostatic skeleton and allows free movement of the visceral organs and for circulation of nutrients.

Eucoelom or true coelom is a fluid-filled cavity that develops within the mesoderm and is lined by mesodermal epithelium called peritoneum. Such animals with a true body cavity are called coelomates or **eucoelomates**. Based on the mode of formation of coelom, the eucoelomates are classified into two types, **Schizocoelomates** – in these animals the

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